PassFrame tuition

Factors, Multiples & Primes

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Find the HCF of 12 and 20.

Method & answer: Factors of 12: 1, 2, 3, 4, 6, 12. Factors of 20: 1, 2, 4, 5, 10, 20. Common factors: 1, 2, 4. HCF = 4.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. List all the factors of 10.

2. List all the factors of 16.

3. Select ALL of these that are multiples of 5: A) 15 B) 18 C) 20 D) 24 E) 25

4. List the first 5 multiples of 7.

5. Is 24 a multiple of 8?

6. Is 6 a factor of 32?

7. Is 23 a prime number? A) Yes B) No, it is divisible by 3 C) No, it is even D) No, it is a square number

8. Is 18 a prime number?

9. List all the factors of 24.

10. What is the smallest multiple of 6 that is greater than 50? A) 48 B) 54 C) 52 D) 60

11. List the prime numbers between 20 and 30.

12. Is 63 a multiple of 9?

13. Which of these is NOT a factor of 30? A) 4 B) 5 C) 6 D) 10

14. What is the first common multiple of 4 and 6 (other than 0)?

15. Is 2 the only even prime number? A) Yes B) No, 4 is also even and prime C) No, all even numbers are prime D) No, there are no even primes

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find all the common factors of 20 and 30.

2. Find the HCF of 18 and 27. A) 3 B) 6 C) 9 D) 18

3. Find the LCM of 5 and 6.

4. Find the LCM of 4 and 9.

6. Is 57 a prime number? A) Yes B) No — $57 = 3 \times 19$ C) No — 57 is even D) No — 57 is a square number

7. Find the first three common multiples of 3 and 4.

8. What is the highest common factor of 15 and 25?

9. Select ALL of these that are square numbers: A) 100 B) 110 C) 121 D) 130 E) 144

11. Find all the factors of 54.

12. Select ALL of these that are prime: A) 41 B) 45 C) 49 D) 53

13. Find the HCF of 8, 12 and 20.

14. A number is a multiple of both 3 and 5. What is the smallest such number? A) 8 B) 10 C) 15 D) 30

5. List all the factor pairs of 40.

10. Is 121 a square number?

15. Find two prime numbers that add to 24.

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. List all the factors of 36 that are also even.

6. A gardener has 48 tulip bulbs and 60 daffodil bulbs. She wants to arrange them into equal-sized rows (same row size for both types) with none left over. What is the greatest possible number of bulbs in each row?

11. A number less than 100 is both a square number and a multiple of 4. List all such numbers.

2. Find the HCF of 24 and 36, then double it.

7. Two bells ring every 6 seconds and every 8 seconds. If they ring together now, how many seconds until they next ring together? A) 14 B) 16 C) 24 D) 48

12. A school has 90 boys and 75 girls. They are split into the largest possible equal-sized groups, with boys and girls in separate groups of the same size, and no pupils left over. How many pupils are in each group, and how many groups are there in total?

3. 5 and 7 are both prime and odd. Give their sum, and say whether the sum of two odd primes is always odd or even.

8. A baker has 84 muffins and 126 cookies. She wants to pack them into boxes so each box has the same number of items and only one type of treat per box, using the largest possible box size. How many items go in each box?

13. A number between 100 and 110 is prime, and is one more than a multiple of 10. Select ALL statements below that are TRUE about this number: A) The number is 101 B) The number is odd C) The number is a multiple of 10 D) The number is prime E) 100 is one of its factors

4. Find the LCM of 6 and 8, then subtract the HCF of 6 and 8 from it. Which is correct? A) 20 B) 22 C) 24 D) 26

9. A number is a multiple of 9, a factor of 108, greater than 20, and odd. What is the number?

14. The HCF of two numbers is 9 and their LCM is 90. One number is 18. Find the other number.

5. A number is prime and also one more than a multiple of 10. Name the smallest such number greater than 10.

10. Three runners complete a lap in 4, 6 and 10 minutes respectively. If they start together, after how many minutes will they all be at the start line together again? A) 20 B) 40 C) 60 D) 120

15. A rectangular garden has an area of 72 m². Its length and width are both whole numbers, and the width is a prime number. Which of these could be the width and length? A) width 2 m, length 36 m B) width 4 m, length 18 m C) width 6 m, length 12 m D) width 8 m, length 9 m